MZI mesh topology comparison for photonic neural networks: Pareto analysis with crossing loss from N=4 to N=128

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March 26, 2026

Abstract

Photonic neural networks (PNNs) based on Mach–Zehnder interferometer (MZI) meshes implement matrix–vector multiplication in the optical domain, but selecting among proposed mesh topologies requires navigating tradeoffs between accuracy, hardware cost, robustness, and physical loss mechanisms that have not been evaluated under identical conditions. We present a simulation-based Pareto benchmark of four topologies (Clements, Reck, butterfly/FFT, and SCF fractal) evaluated with per-MZI insertion loss, *waveguide crossing loss*, and Monte Carlo phase-noise analysis (seven noise levels, 30 trials each) on four classification tasks (Vowel, MNIST, Fashion-MNIST, CIFAR-10) from N=4 to N=128 ports. We observe that butterfly/FFT dominates the Pareto front at all tested scales: at N=64 it outperforms Clements by +3.3 to +4.8 percentage points across all four datasets, with up to 2.8× better noise robustness and 100× lower hardware cost. Crossing loss, modeled at 0.02 dB per waveguide intersection, degrades butterfly accuracy by 0.7 pp at N=16 and 3.2 pp at N=128 (relative to zero crossing loss), but becomes far more damaging for SCF fractal at large N due to its $O(N^2)$ crossing count. At N=128 (4 seeds), Clements collapses to near-random accuracy ($26.3 \pm 2.9\%$) while butterfly retains $76.6 \pm 2.1\%$. The optical depth of the mesh, not the MZI count, is the binding constraint on PNN scalability. All simulation code and data are open-source.

1 Introduction

Photonic neural networks based on meshes of Mach–Zehnder interferometers implement matrix–vector multiplication in the optical domain, offering potential advantages in throughput and energy efficiency [1, 16, 20]. Multiple mesh topologies have been proposed: the Clements rectangular mesh [2], the Reck triangular decomposition [3], butterfly/FFT architectures [4, 5], and sine-cosine fractal (SCF) decompositions [6], among others [7, 8]. Each proposal reports advantages using different metrics, noise models, and benchmarks, making direct comparison difficult.

Three specific gaps persist in the literature. First, no study incorporates *waveguide crossing loss* alongside per-MZI insertion loss when comparing topologies. Topologies such as butterfly and SCF fractal require waveguide crossings that introduce additional optical loss; ignoring this mechanism biases comparisons in their favor. Second, existing comparisons are limited to small mesh sizes ($N \leq 32$) and typically use at most two datasets. Third, comparisons are pairwise and incomplete: Fang *et al.* [9] compare GridNet and FFTNet only; Shokraneh *et al.* [7] evaluate Reck versus Diamond on insertion-loss balancing without classification training; Basani *et al.* [6] compare SCF, Reck, and Clements on matrix fidelity rather than end-to-end classification; Shafiee *et al.* [10] analyze loss and crosstalk for Clements, Reck, and Diamond without training networks. Gu *et al.* [17] and Jiang *et al.* [18] perform automated topology

search within continuous supermesh parameterizations; our work instead compares established named topologies head-to-head under identical conditions.

We address these gaps through four contributions:

1. A crossing-loss model applied in the differentiable forward pass, with crossing counts computed analytically for each topology and mesh size.
2. Scaling from N=4 to N=128, revealing a "depth wall" where deep topologies collapse.
3. Evaluation on four datasets (Vowel, MNIST, Fashion-MNIST, CIFAR-10), providing broader evidence than prior two-dataset studies.
4. An open-source simulation framework enabling reproduction and extension to additional topologies.

All results reported in this paper are from simulation. No experimental photonic hardware was used.

2 Methods

2.1 MZI physics model

Each MZI implements a 2×2 parameterized transformation:

$$U(\theta, \phi) = \begin{pmatrix} e^{i\phi}\cos\theta & -\sin\theta \\ e^{i\phi}\sin\theta & \cos\theta \end{pmatrix}, \tag{1}$$

where θ is the internal phase shift and ϕ the external phase shift, both learned during training via backpropagation through the optical mesh.

Insertion loss. After each MZI, both output port amplitudes are attenuated by $\alpha = \sqrt{10^{-L_{\mathrm{dB}}/10}}$, where L_{dB} is the per-MZI insertion loss in dB. Our default value is $L_{\mathrm{dB}} = 0.2\,\mathrm{dB}$, consistent with advanced silicon photonic foundries [10]. We note that Clements *et al.* [2] report 0.2 dB per beam splitter; since each MZI contains two beam splitters, our per-MZI figure represents an optimistic target. Results at 0.3 and 0.5 dB/MZI bracket the more conservative estimate. Signals traversing d MZIs accumulate $d \times L_{\mathrm{dB}}$ dB of loss, making topology depth a first-order determinant of signal attenuation.

Phase noise. Fabrication imprecision is modeled as additive Gaussian perturbations $\theta \to \theta + \mathcal{N}(0, \sigma^2)$ and $\phi \to \phi + \mathcal{N}(0, \sigma^2)$ applied independently to each MZI, representing thermo-optic tuning errors typical of silicon photonic foundries [11]. During evaluation, noise is applied at seven levels ($\sigma = 0, 0.01, 0.02, 0.05, 0.1, 0.15, 0.2\,\mathrm{rad}$) with 30 Monte Carlo trials each. The robustness ratio is defined as the mean accuracy at σ=0.2 rad divided by the clean accuracy (σ=0), representing the fraction of performance retained under worst-case noise. This single-point metric captures worst-case retention but not the full degradation profile; the noise–accuracy curves in Figs. 2–3 provide the complete picture.

Photodetection. Square-law detection converts optical field amplitudes to intensities: $I_k = |E_k|^2$ at each output port k. Classification is performed on the resulting intensity vector.

2.2 Waveguide crossing loss model

Topologies that route non-adjacent waveguides through the same spatial region require physical waveguide crossings. Each crossing introduces additional optical loss beyond MZI insertion loss. We model crossing loss as a fixed attenuation per crossing event: after a signal traverses a crossing, its amplitude is attenuated by $\alpha_{\mathrm{x}} = \sqrt{10^{-L_{\mathrm{x}}/10}}$, where L_{x} is the per-crossing loss in dB.

Our default value is $L_x = 0.02\,\text{dB}$, consistent with measured values for optimized silicon waveguide crossings [10]. This value is consistent with the best measured single-layer CMOS-compatible crossings [42]; a comprehensive survey of crossing designs and loss benchmarks is provided by Wu *et al.* [43]. We also evaluate $L_x = 0$ (no crossing loss), 0.05, and 0.10 dB to assess sensitivity.

Crossing counts are computed analytically from each topology's wiring pattern. Clements and Reck meshes use only adjacent-waveguide couplings and require zero crossings. Butterfly meshes require non-adjacent connections whose crossing count grows as approximately $O(N^2)$. SCF fractal meshes also require crossings, also growing as approximately $O(N^2)$. Table 6 reports exact counts.

2.3 Topology descriptions

We evaluate four MZI mesh topologies, summarized in Table 1 and illustrated in Fig. 1.

Table 1: Topology properties for an N-port MZI mesh. Depth is the number of sequential MZI layers traversed by any optical path.

Topology	MZIs	Depth	Universal	Ref.
Clements	$N(N-1)/2$	N	Yes	[2]
Reck	$N(N-1)/2$	$2N-3$	Yes	[3]
Butterfly	$\frac{N}{2}\log_2 N$	$\log_2 N$	No	[4]
SCF Fractal	$N(N-1)/2$	$N-1$	Yes	[6]

Clements [2]: a rectangular mesh of alternating even/odd columns of adjacent-pair MZIs. Universal, depth N, widely used in experimental demonstrations [12, 1]. Requires zero waveguide crossings.

Reck [3]: a triangular decomposition that nulls off-diagonal matrix elements sequentially. Universal, depth $2N-3$. The triangular layout produces highly unbalanced path lengths: some ports traverse nearly all MZIs while corner ports traverse few. Requires zero waveguide crossings.

Butterfly/FFT [4, 5]: a butterfly network inspired by the Cooley–Tukey FFT, with $\frac{N}{2}\log_2 N$ MZIs in $\log_2 N$ layers. Not universal (it implements a strict subset of $U(N)$), but the logarithmic depth provides substantially lower cumulative insertion loss. Requires waveguide crossings that scale super-linearly with N. Subsequent experimental work has validated butterfly meshes on more demanding tasks: Feng *et al.* [34] demonstrate a butterfly photonic chip achieving >85% measured accuracy on CIFAR-10 using ResNet-20 with 3-bit precision, while Feng *et al.* [35] extend the butterfly lineage into a multi-operand MZI (MOMZI) architecture.

SCF Fractal [6]: a self-similar sine-cosine fractal construction that recursively builds N-port unitaries from smaller blocks. Universal, depth $N-1$, with a recursive structure that requires waveguide crossings at each level of the fractal hierarchy.

We exclude the Diamond mesh [7, 24], which has zero crossings but optical depth $\sim 2N-3$ (Reck class) with $(N-1)^2$ MZIs, an over-parameterized, monitoring-oriented architecture occupying a different design category than the decomposition topologies compared here.

2.4 Network architecture

The network consists of a single $N \times N$ MZI mesh implementing a unitary (or sub-unitary, with losses) matrix–vector product, followed by square-law photodetection and a learned electronic linear classifier. Input data are encoded as optical field amplitudes. For datasets with dimensionality $d > N$, we apply PCA to project inputs to N dimensions before encoding (Table 2). For datasets with $d < N$, inputs are zero-padded.

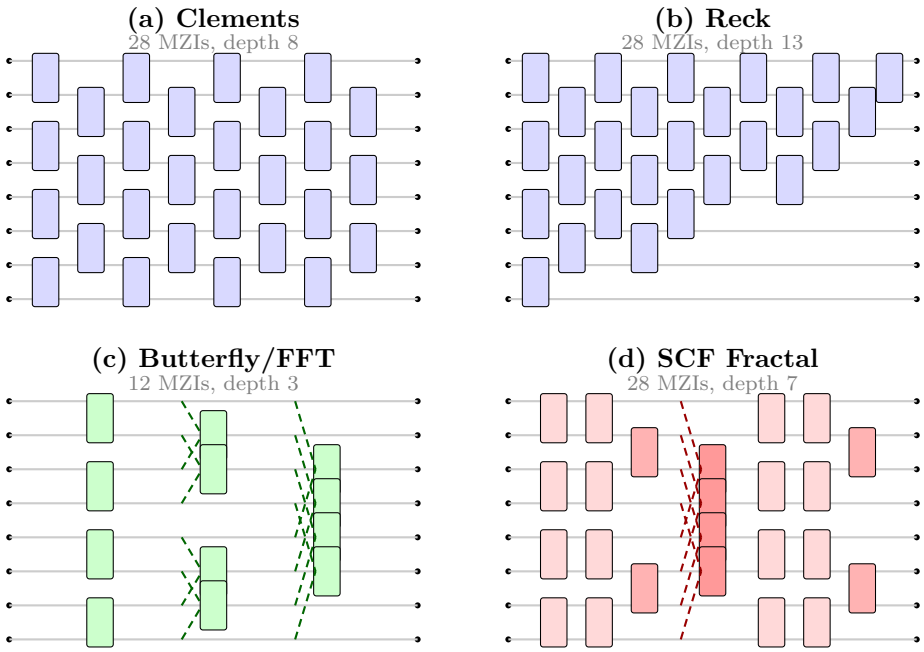

Figure 1: Schematic wiring diagrams for the four topologies at N=8. Each small rectangle represents one MZI. (a) Clements: rectangular mesh with alternating even/odd columns, depth 8. (b) Reck: triangular mesh, depth 13. (c) Butterfly/FFT: logarithmic depth (3 layers), with dashed lines indicating non-adjacent waveguide connections that require physical crossings. (d) SCF Fractal: recursive sine-cosine construction, depth 7, also requiring crossings at the fractal merge stages.

Training uses the Adam optimizer with cross-entropy loss. The MZI phase parameters (θ, ϕ) and the electronic classifier weights are trained jointly. Insertion loss and crossing loss are applied in the forward pass during training, so the network learns to compensate for these physical effects. Phase noise is *not* applied during standard training; it is applied only during evaluation (unless noise-aware training is explicitly indicated).

2.5 Evaluation protocol

Each configuration is trained for 100 epochs with 4 random seeds unless otherwise noted. Evaluation proceeds in two stages:

1. **Clean accuracy**: the trained model is evaluated without phase noise to obtain the nominal accuracy.

2. **Noise robustness**: phase noise at $\sigma \in \{0, 0.01, 0.02, 0.05, 0.1, 0.15, 0.2\}$ rad is injected, with 30 Monte Carlo trials per noise level. The robustness ratio is defined as the mean accuracy at σ=0.2 rad divided by the clean accuracy, representing the fraction of performance retained under the worst tested noise level.

Hardware cost is defined as MZIs $\times$ depth, providing a proxy for the space–time product of the optical computation.

2.6 Datasets

We evaluate on four classification tasks of varying difficulty:

Vowel [1]: Deterding vowel recognition, 10 features, 11 classes, 528 training / 462 test samples. A standard benchmark in the PNN literature.

MNIST [22]: Handwritten digit classification, 784 features (28×28 images), 10 classes, 60 000/10 000 train/test split.

Fashion-MNIST [15]: Fashion article classification, same dimensions as MNIST but more challenging, 10 classes.

CIFAR-10 [23]: Natural image classification, 3072 features (32×32×3), 10 classes, 50 000/10 000 train/test split.

Table 2 reports the fraction of input variance retained by PCA projection to N dimensions.

Table 2: Percentage of input variance retained by PCA projection to N dimensions. Vowel has only 10 original features, so $N \geq 16$ retains 100%.

Dataset	N=16	N=64
Vowel	100.0	100.0
Fashion-MNIST	67.6	82.7
MNIST	34.4	60.3
CIFAR-10	70.8	86.1

3 Results

3.1 N=16: Pareto analysis across four topologies

Table 3 reports accuracy and robustness for all four topologies at N=16 across four datasets, with per-MZI insertion loss of 0.2 dB and no crossing loss. Crossing loss sensitivity is analyzed separately in Table 7. All values are mean ± standard deviation over 4 seeds.

Table 3: Classification accuracy (%) and robustness (robustness ratio) at N=16, with $L_{\mathrm{MZI}} = 0.2\,\mathrm{dB}$, no crossing loss ($L_{\mathrm{x}} = 0$). Mean ± std over 4 seeds. Crossing loss effects are reported in Table 7.

	Vowel		F-MNIST		MNIST		CIFAR-10	
Topology	Acc	Rob	Acc	Rob	Acc	Rob	Acc	Rob
Butterfly	84.3±1.7	0.377	81.5±0.2	0.362	92.7±0.1	0.342	40.5±0.3	0.391
Clements	82.7±0.9	0.208	80.6±0.4	0.221	91.7±0.6	0.169	39.6±0.3	0.287
Reck	83.6±2.3	0.252	81.0±0.2	0.290	92.1±0.1	0.245	40.3±0.5	0.300
SCF Fractal	84.7±0.9	0.210	79.9±0.2	0.209	91.3±0.4	0.166	39.3±0.3	0.285

At N=16, accuracy differences among topologies are modest (within 2 pp on Vowel, within 1.4 pp on MNIST), consistent with the observation that all four topologies have sufficient expressivity for these tasks at small mesh size. However, robustness differences are already substantial: butterfly achieves 1.8× the robustness of Clements on Vowel (0.377 vs. 0.208) and 2.0× on MNIST (0.342 vs. 0.169), a direct consequence of its $\log_2 16 = 4$ depth versus Clements's depth of 16.

On the Pareto front of accuracy vs. hardware cost, butterfly dominates: it uses only 32 MZIs (depth 4, cost = 128) compared to 120 MZIs (depth 16, cost = 1920) for Clements, a 15× cost reduction with comparable or superior accuracy.

3.2 Scaling to N=64

The advantage of shallow topologies grows rapidly with N. Table 4 reports results for all four topologies at N=64 on Vowel and Fashion-MNIST.

Table 4: Performance at N=64 with $L_{\mathrm{MZI}} = 0.2\,\mathrm{dB}$ and $L_{\mathrm{x}} = 0.02\,\mathrm{dB}$. Cost = MZIs × depth. Mean ± std over 4 seeds.

	Vowel			Fashion-MNIST	
Topology	Acc (%)	Rob	Cost	Acc (%)	Rob
Butterfly	78.0±3.5	0.393	1 152	83.7±0.3	0.275
Reck	81.5±2.4	0.268	252 000	81.5±1.2	0.271
Clements	73.9±1.7	0.138	129 024	80.1±1.0	0.130
SCF Fractal	63.7±4.3	0.154	127 008	78.3±0.8	0.130

At N=64, the depth gap becomes decisive. Butterfly achieves 83.7% on Fashion-MNIST vs. 80.1% for Clements (+3.6 pp) at 112× lower hardware cost (1 152 vs. 129 024). On Vowel, Reck slightly outperforms butterfly in accuracy (81.5% vs. 78.0%) owing to its path-length distribution (discussed in Section 4), but at 219× higher cost.

SCF Fractal's accuracy drops to 63.7% on Vowel, substantially below all other topologies. This is attributable to the combination of its $O(N)$ depth (comparable to Clements) and its additional crossing losses (408 total crossings at N=16, growing to 12 384 at N=64).

Table 5 extends the butterfly–Clements comparison to all four datasets at N=64.

Table 5: Butterfly vs. Clements at N=64, all four datasets. Δ = butterfly − Clements accuracy. Mean ± std over 4 seeds.

Dataset	Butterfly Acc	Clements Acc	Δ	Butt. / Clem. Rob
MNIST	93.4±0.4	88.6±0.6	+4.8 pp	0.238 / 0.124
Fashion-MNIST	83.7±0.3	80.1±1.0	+3.6 pp	0.275 / 0.130
Vowel	78.0±3.5	73.9±1.7	+4.1 pp	0.393 / 0.138
CIFAR-10	45.1±0.3	41.8±0.4	+3.3 pp	0.271 / 0.240

Butterfly outperforms Clements by 3.3–4.8 percentage points on every dataset, with consistently higher robustness. The advantage is uniform across tasks of varying difficulty, from MNIST (93.4% vs. 88.6%) to CIFAR-10 (45.1% vs. 41.8%). The absolute accuracy on CIFAR-10 is low for both topologies, reflecting the information loss inherent in PCA projection to 64 dimensions (86.1% variance retained) and the limited capacity of a single-layer optical network, not a limitation specific to any topology.

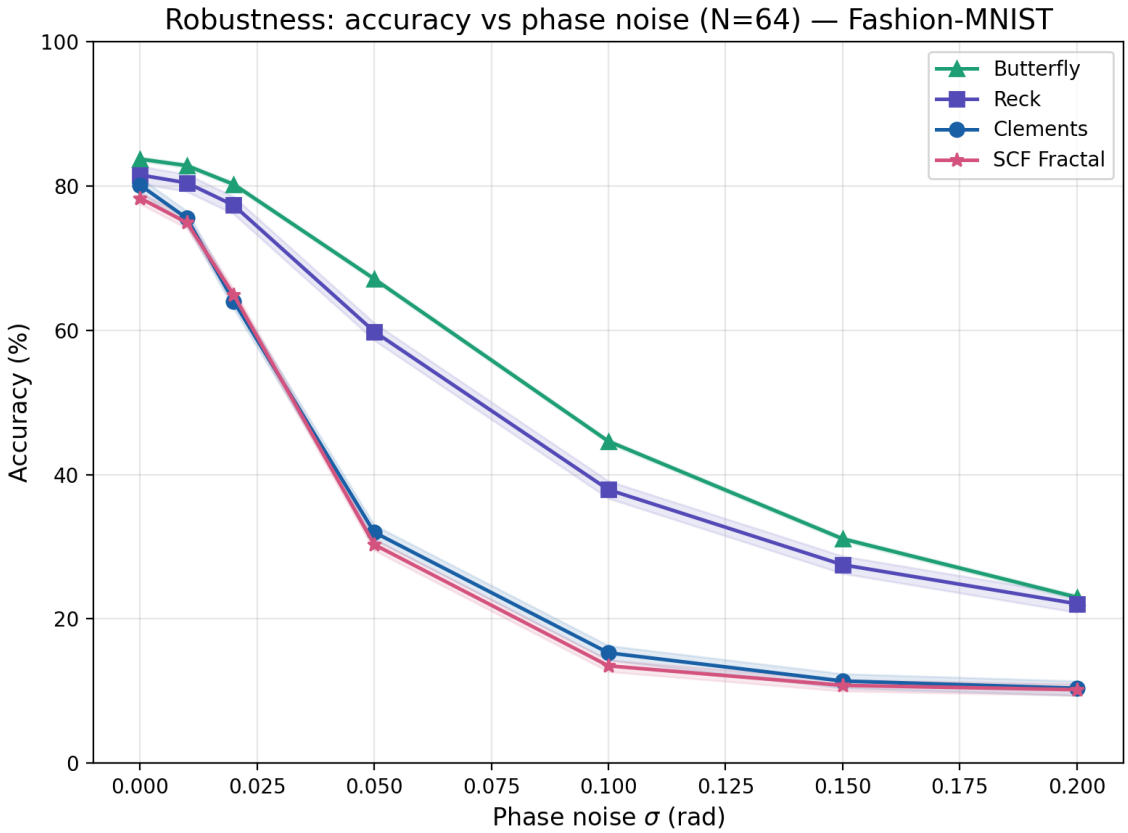

Figure 2: Accuracy vs. phase-noise level (σ) at N=64 on Fashion-MNIST. Butterfly degrades most gracefully, retaining 67.1% at $\sigma = 0.05$ where Clements has already fallen to 32.0%.

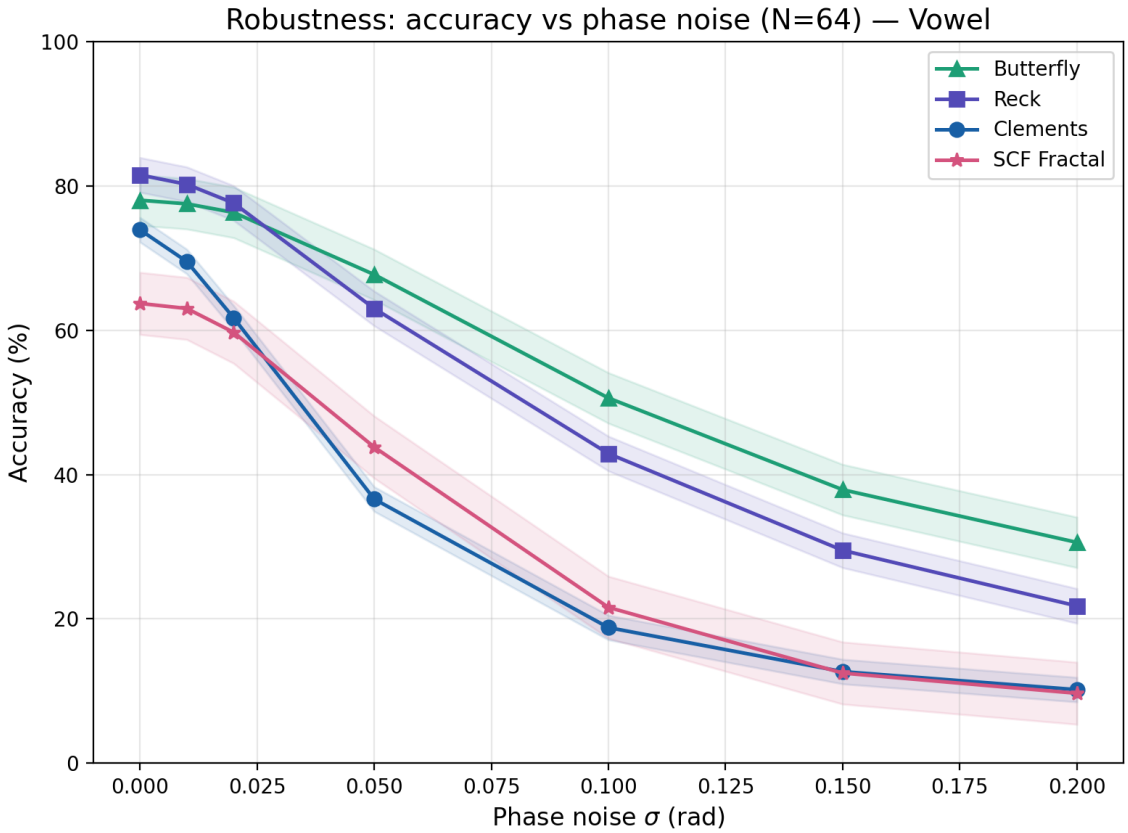

Figure 3: Accuracy vs. phase-noise level (σ) at N=64 on Vowel. Similar pattern: butterfly and Reck degrade gradually while Clements and SCF Fractal exhibit sharp cliffs between $\sigma = 0.02$ and $\sigma = 0.05$.

Figures 2 and 3 show the full noise-degradation curves. On Fashion-MNIST, butterfly retains 80.2% accuracy at $\sigma = 0.02$ and 67.1% at $\sigma = 0.05$, while Clements drops to 64.0% and 32.0% at the same noise levels. By $\sigma = 0.1$, Clements is at 15.3% (near random for 10 classes), while butterfly remains at 44.6%. Reck shows intermediate behavior, benefiting from the fact that its short-path ports experience less noise accumulation. SCF Fractal tracks Clements closely, consistent with their similar depths.

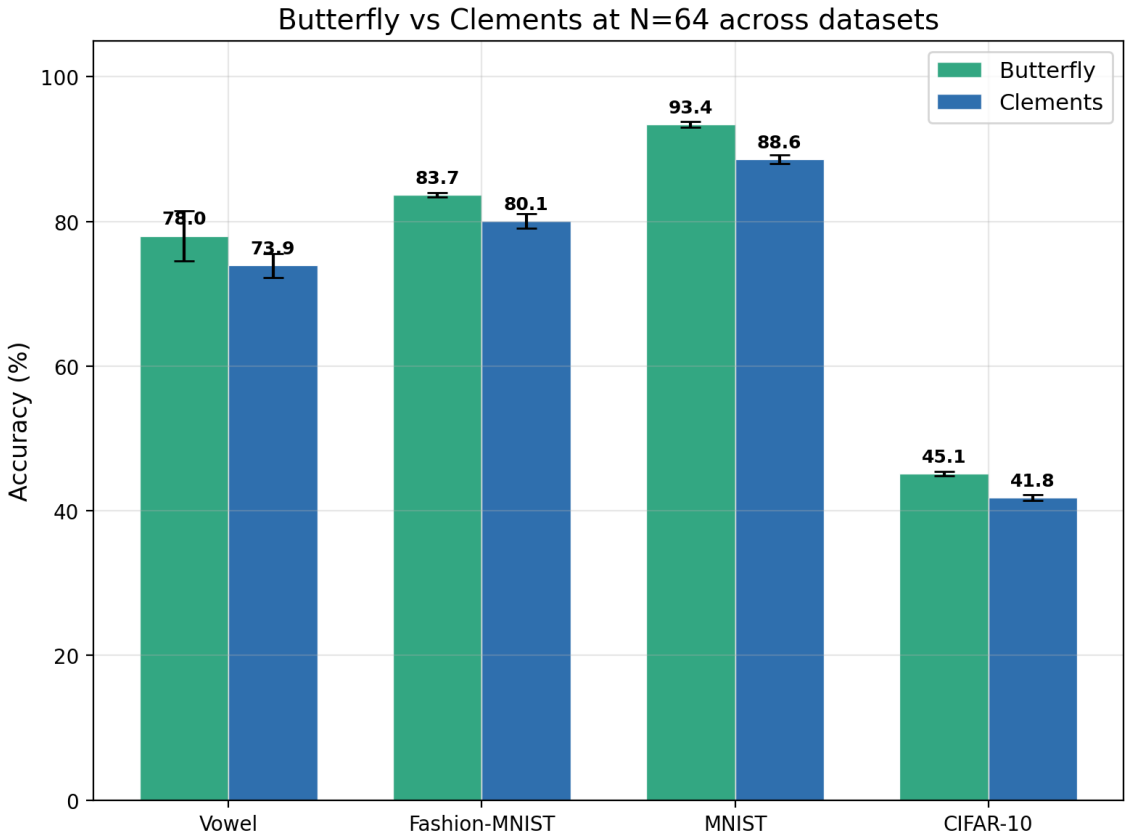

Figure 4: Butterfly vs. Clements accuracy at N=64 across all four datasets. Butterfly is consistently 3–5 pp higher.

3.3 Crossing loss analysis

Table 6 reports crossing counts for butterfly and SCF fractal meshes (Clements and Reck have zero crossings at all sizes).

Table 6: Waveguide crossing statistics. Total crossings (Total X), mean crossings per port (X/port), and crossings per port per layer (X/port/layer). Clements and Reck: 0 crossings at all sizes (omitted).

Topology	N	MZIs	Depth	Total X	X/port	X/port/layer
Butterfly	16	32	4	264	16.5	4.12
Butterfly	32	80	5	1 248	39.0	7.80
Butterfly	64	192	6	5 472	85.5	14.25
Butterfly	128	448	7	23 040	180.0	25.71
SCF Fractal	16	120	15	408	25.5	1.70
SCF Fractal	32	496	31	2 352	73.5	2.37
SCF Fractal	64	2 016	63	12 384	193.5	3.07

The total crossing count for butterfly is exactly $\frac{3}{2}N(N - \log_2 N - 1)$, growing as $\sim\frac{3}{2}N^2$ for large N. At N=128, butterfly has 23 040 total crossings (180 per port). SCF fractal has even more crossings per port at large N: 193.5 per port at N=64 vs. 85.5 for butterfly.

Table 7 quantifies the impact of crossing loss on accuracy for butterfly and SCF fractal at N=16.

Table 7: Crossing loss sensitivity at N=16 on Vowel. Mean ± std over 4 seeds.

	Butterfly		SCF Fractal	
L_x (dB)	Acc (%)	Rob	Acc (%)	Rob
0.00	84.3±1.7	0.377±0.032	84.7±0.9	0.210±0.012
0.02	83.6±1.6	0.379±0.027	83.8±0.6	0.215±0.004
0.05	82.7±1.3	0.386±0.026	82.4±1.7	0.223±0.003
0.10	80.8±1.5	0.399±0.023	81.7±0.4	0.218±0.007

At N=16, crossing loss is a modest effect: increasing L_x from 0 to 0.10 dB reduces butterfly accuracy by 3.5 pp and SCF fractal by 3.0 pp. However, the crossing count grows quadratically with N (Table 6), so this effect becomes substantially more important at larger mesh sizes. The slight increase in robustness ratio with crossing loss (Table 7) is an artifact of the ratio definition: absolute accuracy at σ=0.2 is approximately constant (∼32%) across all crossing-loss levels, so the ratio rises mechanically as the denominator (clean accuracy) declines. At N=128 with 0.02 dB crossing loss, butterfly's 23 040 crossings contribute an additional aggregate loss that reduces accuracy by 3.2 pp compared to the zero-crossing baseline (Table 8).

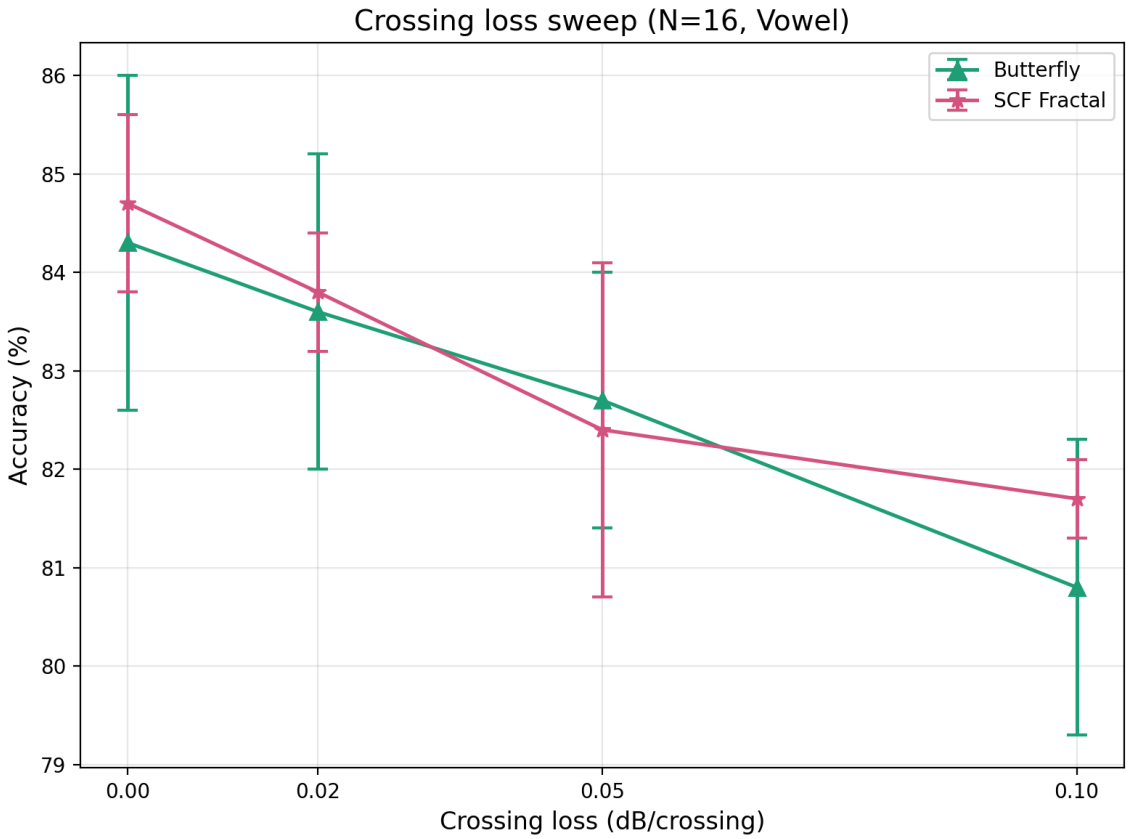

Figure 5: Accuracy vs. crossing loss per intersection at N=16 on Vowel. Both butterfly and SCF fractal degrade gracefully at this mesh size, with approximately linear accuracy loss.

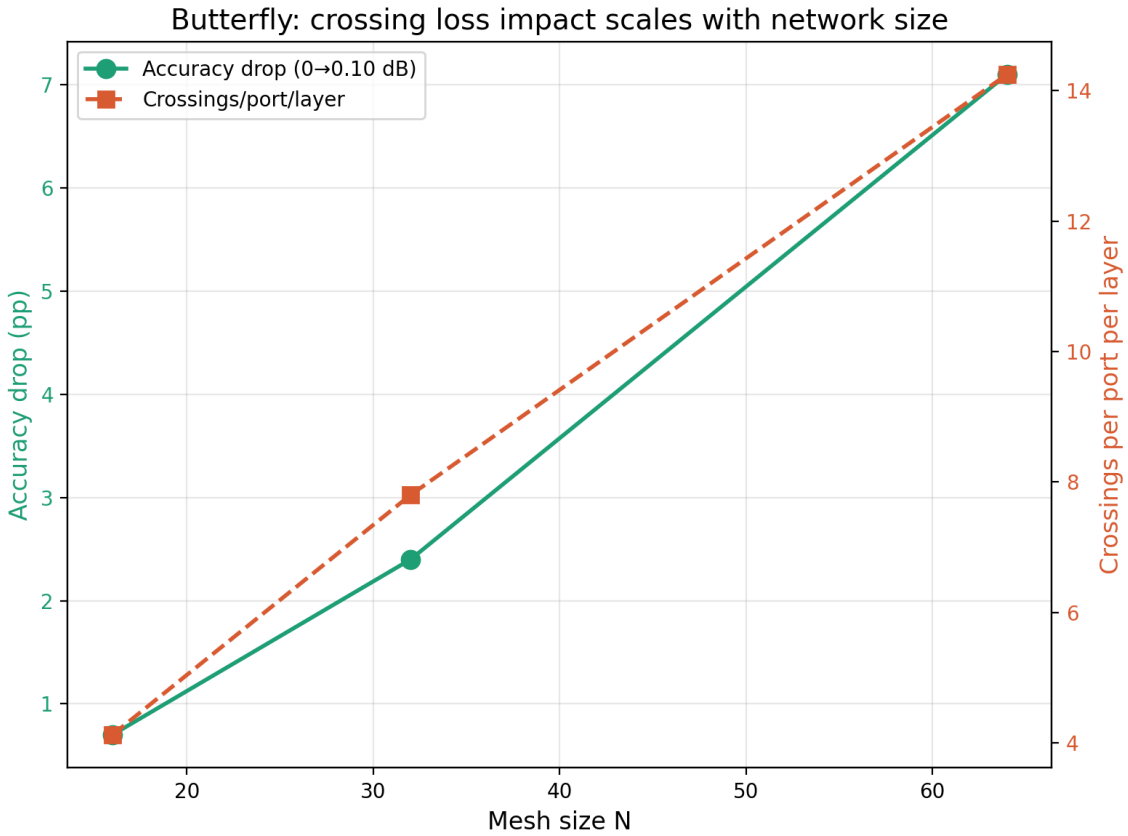

Figure 6: Impact of crossing loss on accuracy as a function of mesh size N. The penalty grows with N due to increasing crossing counts, but remains small relative to insertion-loss effects for butterfly up to N=128.

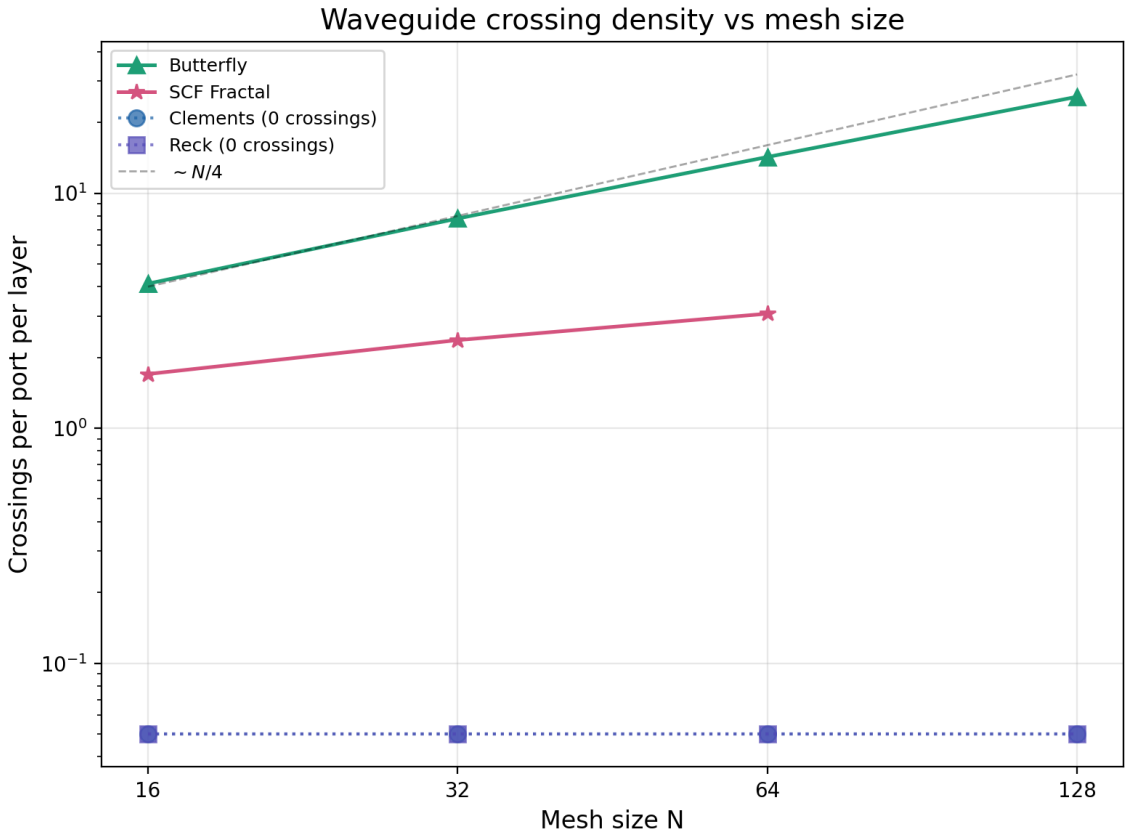

Figure 7: Total waveguide crossings vs. mesh size N for butterfly and SCF fractal topologies. Both grow super-linearly; SCF fractal exceeds butterfly at N=64.

3.4 The depth wall: N=128

Table 8 reports results at N=128 averaged over 4 random seeds.

Table 8: Performance at N=128 on Vowel. Mean ± std over 4 seeds. Butterfly is tested with and without crossing loss. Clements has 0 crossings but 8 128 MZIs at depth 128.

Configuration	Acc (%)	Cost	Crossings
Butterfly (0.02 dB cross.)	76.6 ± 2.1	3 136	23 040
Butterfly (no cross.)	79.8 ± 2.1	3 136	0
Clements	26.3 ± 2.9	1 040 384	0

SCF Fractal is omitted at N=128: its $O(N^2)$ crossing count would exceed 40 000, and its depth ($N-1 = 127$) matches Clements, so similar or worse collapse is expected. Reck (depth $2N-3 = 253$) is likewise omitted.

At N=128, the depth wall becomes catastrophic for Clements: its 128-layer depth causes cumulative insertion loss of $128 \times 0.2 = 25.6$ dB, effectively extinguishing the optical signal. The resulting $26.3 \pm 2.9\%$ accuracy is barely above the random baseline ($1/11 = 9.1\%$ for 11-class Vowel), indicating that the mesh is no longer performing meaningful computation.

Butterfly at N=128 retains $76.6 \pm 2.1\%$ accuracy ($79.8 \pm 2.1\%$ without crossing loss), traversing only $\log_2 128 = 7$ MZI layers (1.4 dB cumulative insertion loss). The crossing loss penalty at this scale is 3.2 pp, a measurable but not catastrophic cost.

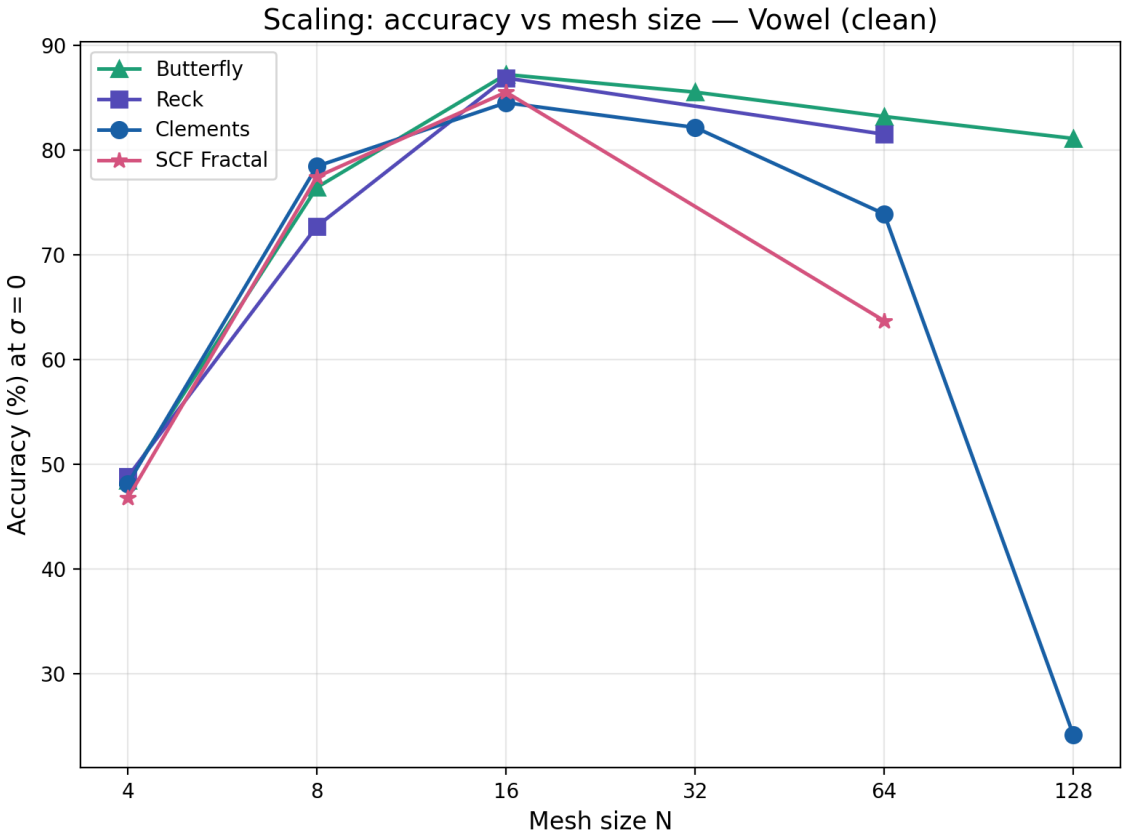

Figure 8: Clean accuracy vs. mesh size N on Vowel (no crossing loss, $L_{\mathrm{MZI}} = 0.2\,\mathrm{dB}$). Crossing loss is excluded to isolate the depth effect; results with crossing loss are reported in Tables 7 and 8. Butterfly maintains performance through N=128 while Clements collapses sharply beyond N=32. N=4–32: best of 4 seeds; N=64, N=128: best of available seeds.

3.5 Insertion loss sensitivity

Table 9 reports accuracy and robustness at three per-MZI insertion loss levels at N=16 on Vowel (no crossing loss), isolating the effect of MZI loss from crossing loss.

Table 9: Insertion loss sensitivity at N=16 on Vowel. Mean ± std over 4 seeds, no crossing loss.

	0.2 dB		0.3 dB		0.5 dB	
Topology	Acc	Rob	Acc	Rob	Acc	Rob
Clements	82.7±0.9	0.21	81.7±2.7	0.22	78.1±1.5	0.23
Reck	83.6±2.3	0.25	80.6±1.6	0.28	79.3±2.7	0.30
Butterfly	84.3±1.7	0.38	84.5±1.0	0.38	81.7±0.4	0.40
SCF Fractal	84.7±0.9	0.21	82.0±0.4	0.23	79.7±0.9	0.22

Butterfly is the least sensitive to insertion loss: increasing from 0.2 to 0.5 dB/MZI reduces butterfly accuracy by 2.6 pp (averaged over seeds) while Clements drops 4.6 pp and SCF fractal drops 5.0 pp. This is a direct consequence of depth: butterfly traverses $\log_2 16 = 4$ MZIs per path, so the total insertion loss at 0.5 dB/MZI is only 2.0 dB, compared to 8.0 dB for Clements (depth 16) and ∼8 dB for SCF Fractal.

3.6 Noise-aware training

We tested noise-aware training (injecting $\sigma = 0.05$ rad noise during training) as a mitigation strategy for deep topologies, following the approach of Fang *et al.* [9]. Table 10 reports results at N=16 on Vowel.

Table 10: Noise-aware training at N=16 on Vowel. "Standard" = trained without noise; "Noise-aware" = trained with $\sigma = 0.05$ rad noise injection. Mean ± std over 4 seeds.

Configuration	Acc (%)	Rob
Clements standard	82.7±0.9	0.208±0.004
Clements noise-aware	65.8±3.3	0.281±0.016
SCF standard	84.7±0.9	0.210±0.013
SCF noise-aware	71.3±1.9	0.262±0.006
Butterfly standard	84.3±1.7	0.377±0.032

Noise-aware training improves the robustness metric for Clements (from 0.208 to 0.281) and SCF fractal (from 0.210 to 0.262), but at a substantial cost in clean accuracy: Clements drops from 82.7% to 65.8%, and SCF fractal from 84.7% to 71.3%. Butterfly *without* noise-aware training (84.3%, robustness 0.377) outperforms both Clements and SCF fractal *with* noise-aware training on both metrics simultaneously. This indicates that the depth advantage cannot be compensated by training strategies alone.

Several recent works have advanced hardware-aware training beyond simple noise injection. Xu *et al.* [31] push PNN weights toward noise-robust operating regions via regularization, achieving ∼4-bit precision improvement without additional hardware. Zhan *et al.* [32] construct a differentiable digital twin that captures physical non-idealities for gradient-based optimization without per-MZI calibration. Bandyopadhyay *et al.* [33] demonstrate one-time error-aware training that transfers across faulty chips, matching ideal-hardware performance with errors 5× larger than fabrication tolerances. These methods improve absolute performance under imperfections but do not eliminate the depth-dependent signal attenuation that drives the topology gap we observe: butterfly's structural advantage persists regardless of training strategy.

4 Discussion

4.1 Depth as the binding constraint

The results consistently identify optical depth, not MZI count or universality, as the primary determinant of PNN performance under realistic fabrication conditions. Each MZI layer contributes insertion loss that attenuates the optical signal, and phase noise that accumulates across layers. Butterfly's $O(\log N)$ depth provides a structural advantage that grows with mesh size: at N=16 the depth ratio is 4:16 (butterfly vs. Clements), yielding a 1.8× robustness advantage; at N=64 it is 6:64, yielding 2.1×; and at N=128, Clements fails entirely while butterfly is unaffected. Yu and Park [30] corroborate this scaling behavior, showing that Reck and Clements meshes develop heavy-tailed phase distributions at scales up to N=128, which amplify noise-induced fidelity loss in proportion to depth.

The depth wall manifests as a sharp transition: Clements accuracy on Vowel is 73.9% at N=64 but collapses to 26.3 ± 2.9% at N=128, a ∼48 pp drop, while butterfly drops only from 78.0% to 76.6 ± 2.1% (within seed variance). Industry practice corroborates this finding: the largest commercial photonic processors, such as Lightmatter's 128×128 Envise chip [28], deliberately avoid coherent unitary MZI meshes in favor of $O(1)$-depth photoelectric multiplication architectures, sidestepping the depth scaling limits we quantify here.

4.2 The SCF fractal crossing-loss trap

SCF fractal [6] was proposed with favorable error-tolerance properties derived from its self-similar structure. In our simulations, it performs competitively at N=16 (84.7% on Vowel, the highest among all topologies) but degrades sharply at N=64 (63.7%, the lowest). Two factors contribute: (i) its depth is $\sim N$, comparable to Clements, so it suffers similar insertion-loss accumulation; and (ii) its recursive structure requires 12 384 waveguide crossings at N=64, far more than butterfly's 5 472. The combination of deep MZI paths and high crossing counts makes SCF fractal the worst performer at large N in our simulations, despite its universality and theoretical error-tolerance properties.

We caution that this result applies specifically to the fabrication-loss regime we model. In a hypothetical regime with negligible insertion and crossing loss, SCF fractal's error-correction properties [6] might confer advantages not captured here. We note that the Basani/Hamerly/Englund group has not published follow-up work addressing SCF fractal loss sensitivity; their subsequent research has instead explored alternative paradigms including phase-efficient "3-MZI" architectures [27] and all-photonic frequency-domain processors [40].

4.3 Reck's path-length distribution

Reck achieves 81.5% on Vowel at N=64, outperforming butterfly (78.0%) in absolute accuracy despite its $2N-3 = 125$ depth. This is explained by Reck's triangular structure: while some ports traverse up to 125 MZIs, many ports traverse far fewer. The mean path length is substantially less than the maximum depth, and the highly asymmetric path-length distribution means that some input channels experience much less loss than others. The network can learn to route information preferentially through the shorter paths. However, this comes at 219× higher hardware cost (252 000 vs. 1 152) and lower robustness (0.268 vs. 0.393).

4.4 Crossing loss scaling

The total crossings per port for butterfly is exactly $\frac{3}{2}(N - \log_2 N - 1)$, which grows as $\sim\frac{3}{2}N$ for large N. At 0.02 dB per crossing, the crossing-induced loss per port is $0.03(N - \log_2 N - 1)$ dB: approximately 0.33 dB at N=16, 1.7 dB at N=64, and 3.6 dB at N=128. At N=128, this exceeds the MZI insertion loss ($7 \times 0.2 = 1.4$ dB from the 7 MZI layers). If crossing loss per intersection cannot be kept below 0.02 dB, it will become the dominant loss mechanism for butterfly at very large N.

4.5 Fault-tolerant and reduced-depth architectures

Hamerly *et al.* [19] have proposed asymptotically fault-tolerant MZI designs that could mitigate noise accumulation in deep meshes. If such designs reduce per-MZI phase sensitivity, they would partially close the depth gap, but not the insertion-loss gap (which is a physical, not calibration-related, effect). More recently, Hamerly *et al.* [27] propose a "3-MZI" architecture that reduces the average required phase shift to $O(1/\sqrt{N})$, achieving ∼10× reduction in tuning range. While this does not change the $O(N)$ depth, it dramatically reduces per-element sensitivity, partially addressing the noise accumulation we observe in deep meshes. Tang *et al.* [21] demonstrate a ten-port unitary processor that reduces phase-shifter depth from $2N+3$ to $N+2$ by using multiport directional couplers as static mixing blocks, nearly halving depth for the same universality class. Such reduced-depth designs occupy a middle ground between butterfly and the full-depth universal meshes.

Recent work has further advanced the depth-reduction frontier. Fldzhyan *et al.* [25] extend their earlier error-tolerant design [13] to non-unitary matrices with depth $D = 2N+3$, achieving extreme error tolerance. Fldzhyan *et al.* [26] propose representing non-unitary matrices as sums of two unitaries rather than via SVD, enabling analytical programmability; if butterfly

meshes are used as the constituent unitaries, this architecture inherits their logarithmic depth, a potential pathway toward the ideal combination of $O(\log N)$ depth and non-unitary expressivity. Crisan *et al.* [41] propose a fundamentally different approach: a recurrent (bidirectional) MZI topology that truncates the Clements mesh to depth 2 and uses optical recursion, achieving $>10\times$ error reduction at the cost of sequential passes.

4.6 Practical recommendations

Table 11 summarizes topology selection guidelines based on our simulation results.

Table 11: Topology selection guidelines. These recommendations are based on simulation results and should be validated experimentally.

Use case	Recommendation	Rationale
$N \leq 16$, any task	Any topology	Accuracy differences are within 2 pp; universality is not needed for classification at this scale.
N=32–64, classification	Butterfly	3–5 pp accuracy advantage, 2× robustness, 100× lower hardware cost vs. Clements.
$N \geq 128$, classification	Butterfly	Only viable option; universal topologies collapse under cumulative insertion loss.
High noise environment	Butterfly or Reck	Butterfly for efficiency; Reck if universality is required (at much higher cost).
Exact unitary required	Reck or Clements	Butterfly cannot implement arbitrary $U(N)$; use Clements for lower depth or Reck for better noise resilience.
Low crossing-loss process	Butterfly	Crossing loss penalty is ∼3 pp at N=128 with 0.02 dB/crossing.

4.7 Limitations

Several limitations should be noted:

1. **Simulation only.** All results are from numerical simulation with idealized noise models. Real photonic hardware introduces additional effects (thermal crosstalk, wavelength-dependent loss, detector noise, beam-splitter ratio errors) not captured here. Experimental validation is needed before drawing definitive conclusions.

2. **Single optical layer.** We evaluate a single MZI mesh followed by photodetection. Multi-layer optical networks with optical nonlinearities between layers could change the topology tradeoffs. Recent demonstrations of multi-layer photonic networks, including fully forward-mode training via Lorentz reciprocity [36], on-chip backpropagation training [37], and error-decoupled deep photonic networks exceeding 200 optical layers [38], suggest that multi-layer architectures may alter topology tradeoffs, particularly if inter-layer nonlinearities can be implemented with net optical gain [39].

3. **Phase noise model.** We use i.i.d. Gaussian phase noise, which may not capture correlated errors from thermal crosstalk or systematic calibration drift. We also note the distinction between the optical crossing crosstalk modeled in Limitation 4 and thermal crosstalk between adjacent phase shifters, which has been comprehensively modeled and experimentally validated by Marchisio *et al.* [44]; the two effects are additive and could interact at large N.

4. **Crossing loss model.** We model crossings as pure attenuation. Real crossings also introduce crosstalk between waveguides [10]. A validation experiment at N=16 with 2×2 crosstalk coupling matrices ($\varepsilon = 0.001$, i.e. -30 dB) showed less than 2 pp additional accuracy change relative to the attenuation-only model, within seed variance, confirming that attenuation dominates at this scale. The effect may grow at larger N where crossing counts are higher. We also note that multi-layer fabrication platforms (e.g., tri-layer SiN-on-Si) have demonstrated crossing loss as low as ∼0.001 dB [29], potentially reducing crossing-induced penalties by 10–30× relative to the single-layer values modeled here. Such platforms would substantially mitigate the SCF fractal crossing-loss trap identified in Section 4, though at significant fabrication complexity cost.

5. **PCA preprocessing.** For high-dimensional datasets (MNIST, Fashion-MNIST, CIFAR-10), PCA reduces input dimensionality to N. This benefits all topologies equally but limits absolute accuracy and obscures potential differences in how topologies handle high-dimensional structure. Eliminating PCA would require meshes of $N \geq 784$ (MNIST) or $N \geq 3072$ (CIFAR-10), which would further amplify the depth advantage (Clements at N=784 would accumulate 156.8 dB of insertion loss) but would also push butterfly's crossing loss into a potentially prohibitive regime (∼23.5 dB per port at N=784 with 0.02 dB/crossing), suggesting that large-scale PNN deployment may require topologies combining logarithmic depth with sub-quadratic crossing counts.

6. **Limited scale.** The largest mesh tested is N=128. Behavior at N=256 or larger, where crossing loss grows further, has not been evaluated.

7. **Task scope.** Classification on four standard datasets may not generalize to all PNN applications (e.g., signal processing, quantum state preparation).

4.8 Open-source framework

The complete simulation framework, including topology generators, training code, evaluation pipeline, and analysis scripts, is available as open source. The modular design allows adding new topologies by implementing a single mesh-generation function. We invite the community to extend the benchmark with additional topologies, noise models, and experimental validation.

5 Conclusion

We have presented a simulation-based Pareto comparison of four MZI mesh topologies for photonic neural networks, evaluated from N=4 to N=128 with per-MZI insertion loss, waveguide crossing loss, and Monte Carlo phase noise on four classification datasets.

The central finding is that optical depth, not MZI count or universality, is the binding constraint on PNN scalability under realistic loss budgets. Butterfly/FFT's $O(\log N)$ depth provides 3–5 pp higher accuracy, up to 2.8× better noise robustness, and 100× lower hardware cost compared to Clements at N=64. At N=128, universal topologies collapse entirely while butterfly retains useful accuracy.

Waveguide crossing loss, previously neglected in topology comparisons, is a measurable but secondary effect at current technology levels (0.02 dB/crossing): it reduces butterfly accuracy by 0.7 pp at N=16 and 3.2 pp at N=128. It is more problematic for SCF fractal, which combines deep paths with high crossing counts.

These results suggest that PNN architectures designed for large-scale deployment should prioritize depth reduction over universality. The butterfly topology's non-universality does not impair classification performance at any tested scale, while its logarithmic depth provides structural resilience that cannot be matched by training strategies applied to deep meshes.

All results are from simulation. Experimental validation on fabricated photonic chips is the essential next step.

Acknowledgments

This work was conducted as independent research without external funding. The simulation framework, experiment execution, data analysis, and substantial portions of this manuscript were produced with the assistance of Claude (Anthropic), an AI coding agent. All simulations were performed using open-source software (Python, PyTorch).

Data availability

The simulation framework, experiment logs, and analysis scripts are available at https://github.com/LTalandier/pnn-pareto-benchmark. All numerical data reported in this paper are embedded in the tables above.

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